M101 · Unit 5

Derivatives and differentiation methods

Recall: retrieve readily · Understand: explain conditions · Consult: look up while working

Use	Topic	Statement and conditions
Understand	**Derivative**	$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$ Require a finite limit from both sides at an interior point and $f(a)$ defined. At an endpoint specify an inward derivative. Differentiability implies continuity, not conversely.
Recall	**Meaning and notation**	$f'(x) = \mathrm{d}f \,/\, \mathrm{d}x$: instantaneous output/input rate. Tangent: $y - f(a) = f'(a)(x - a)$. If output has units U and input X, derivative units are $U \,/\, X$.
Recall	**Linear combinations**	$(c)' = 0$, $(cf + dg)' = cf' + dg'$ for constants c, d and differentiable functions f and g.
Recall	**Powers**	$(x^n)' = nx^{n-1}$. Integer powers: all real inputs except $x = 0$ for negative powers; a constant is differentiated directly. Real powers: use $x > 0$; investigate any additional real-domain points separately. $(\sqrt{x}\,)' = 1 \,/\, (2\sqrt{x}\,)$ for $x > 0$.
Recall	**Exponential and log**	$(e^x)' = e^x$; $(a^x)' = a^x \ln a$ for $a > 0$. $(\ln x)' = 1 \,/\, x$ for $x > 0$; $(\log_a x)' = 1 \,/\, (x \ln a)$ for $a > 0$, $a \neq 1$, $x > 0$.
Recall	**Trigonometric**	$(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\tan x)' = 1 \,/\, \cos^2 x$ where $\cos x \neq 0$. Arguments in radians.
Recall	**Product and quotient**	$$(fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$ Both factors differentiable; quotient requires $g \neq 0$.
Understand	**Chain rule**	$$\frac{\mathrm{d}}{\mathrm{d}x} F(g(x)) = F'(g(x))g'(x).$$ Require g differentiable at x and F at $g(x)$. Evaluate each derivative at its own input. Retain every layer and its domain.

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Branches, further rates and checks

Use	Topic	Statement and conditions
Consult	**Implicit derivatives**	On a differentiable branch $y = y(x)$, $(y^n)' = ny^{n-1}y'$, $(xy)' = y + xy'$. Collect y' terms; divide only by a nonzero coefficient. Verify the point on the original relation. An exceptional denominator needs investigation.
Understand	**Second derivative**	$f'' = (f')' = \mathrm{d}^2 f / \mathrm{d}x^2$, not $(f')^2$. Units U / X^2. Keep y dependent when differentiating implicitly again. Substitute a point after differentiating.
Consult	**Joins and exclusions**	For smoothly extendible branch formulas, first match function values, then one-sided slopes. Matching first derivatives need not give a second derivative. Cancellation does not fill a hole in the original domain.
Understand	**Checking**	Use a different algebraic route, units and varied inputs. A finite secant $(f(x+h) - f(x)) / h$ is not generally an exact derivative. Both offset signs matter. Finite agreement is evidence, not proof; tiny offsets can cause rounding problems.

Consult · Python quick lookup

```
from math import exp, isfinite
x, h = 1.0, 0.01  # dimensionless test input and offset
if not isfinite(x) or not isfinite(h):
    raise ValueError("Use finite inputs")
if h == 0 or x + h == x:
    raise ValueError("Choose a representable nonzero offset")
f0, f1 = x*x*exp(-x), (x+h)**2*exp(-(x+h))
print("secant", (f1-f0)/h)
print("derivative", exp(-x)*(2*x-x*x))
```

This example uses the stated safe inputs. For changed functions or extreme inputs, check domains, finite outputs and overflow as in §5.27. Repeat both offset signs at several inputs.

Notation: primes: derivatives with respect to the displayed input; h: nonzero offset; c, d, a, n: constants in their stated formulas. Retain physical units; exponential, logarithm and trigonometric arguments are dimensionless.

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